UNIT QUESTIONS

SKILLS CRITICAL THINKING (5)

1 **a** Which of the following does not move as a transverse wave?

A ultraviolet light

B sound wave

C surface water wave

D microwave

(1)

b Which of these effects describes the change in pitch we hear when a fast moving motorbike goes past?

A total internal reflection

B refraction

C reflection

D Doppler

(1)

(6)

c Which of the following does not make use of total internal reflection?

A oscilloscope

B endoscope

C prismatic periscope

D optical fibres

(1)

(Total for Question 1 = 3 marks)

SKILLS INTERPRETATION

2 The diagram below shows the cross-section of a water wave.



a Copy this diagram and mark on it:

(5)

i the wavelength of the wave (λ) **(1)**

ii the amplitude of the wave (A). **(1)**

SKILLS PROBLEM SOLVING (8)

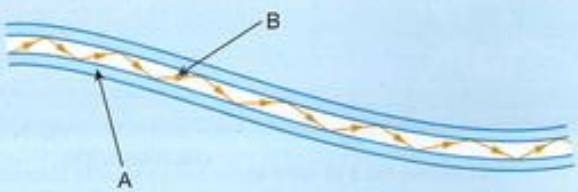
b **i** A water wave travelling at 20 m/s has a wavelength of 2.5 m. Calculate the frequency of the wave. **(3)**

ii Calculate the time period of the above wave. **(1)**

(Total for Question 2 = 6 marks)

SKILLS ANALYSIS

3 The diagram below shows a ray of light travelling down an optical fibre.



a What is A? **(1)**

b What is B? **(1)**

c Why is light reflected from the boundary between A and B? **(2)**

SKILLS CRITICAL THINKING

d Describe one medical use for optical fibres. **(1)**

(Total for Question 3 = 5 marks)

4

a i Explain the difference between a longitudinal wave and a transverse wave. **(1)**

ii Give one example of each type of wave. **(2)**

A girl stands 500 m from a tall building and bangs two pieces of wood together. At the same moment her friend starts a stopwatch. The sound waves created by the two pieces of wood hit the building and are reflected. When the two girls hear the echo they stop the stopwatch and note the time. The girls repeat the experiment four more times. The results are shown in the table below.

Experiment	Time (seconds)
1	2.95
2	3.00
3	2.90
4	3.20
5	2.95

SKILLS EXECUTIVE FUNCTION

b Why did the girls repeat the experiment five times? **(1)**

SKILLS PROBLEM SOLVING

c Calculate the speed of sound using the results. **(6)**

SKILLS EXECUTIVE FUNCTION

d One of the girls thought that their answer might be affected by wind. Was she correct? Explain your answer. **(2)**

(Total for Question 4 = 12 marks)

SKILLS CRITICAL THINKING

5

The electromagnetic spectrum contains the following groups of waves: infrared, ultraviolet, x-rays, radio waves, microwaves, visible spectrum and gamma rays.

a Put these groups of waves in the order they appear in the electromagnetic spectrum starting with the group that has the longest wavelength. **(2)**

b Write down four properties that all of these waves have in common. **(4)**

c Write down one use for each group of waves. **(7)**

d Which three groups of waves could cause cancer? **(3)**

e Which three groups of waves can be used to communicate **(3)**

(Total for Question 5 = 19 marks)

SKILLS PROBLEM SOLVING

6

a A ray of light hits the outside surface of a glass block with an angle of incidence of 38°. Its angle of refraction inside the block is 24°.

i Calculate the refractive index of the glass. **(4)**

ii Calculate the critical angle for this glass. **(2)**

b Calculate the refractive index for a piece of glass whose critical angle is 42°. **(3)**

(Total for Question 6 = 9 marks)